

Supporting Information for:
Two-dimensional powder diffraction in layered van der Waals films:
quantitative X-ray area-detector modeling

This Supporting Information follows the physical and experimental sequence of the main manuscript. Sections S1–S5 develop the diffraction model and its detector observable. Section S6 covers the ordered-film comparisons, followed by the PbI_2 extension in S7 and calibration, refinement, and uncertainty in S8.

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S1 Incident beam and scattering geometry

This section supports the main-text scattering geometry. An incident ray first reaches the sample, and an accepted scattered ray then reaches the active detector. The coordinate definitions below keep those two restrictions separate.

The notation follows the main manuscript. Bold symbols denote vectors or explicitly identified matrices; a hat on a vector denotes unit magnitude, and T denotes transpose. The imaginary unit is i , and a superscript $*$ denotes complex conjugation except on the reciprocal basis vectors $\mathbf{a}^*, \mathbf{b}^*, \mathbf{c}^*$. Descriptive subscripts such as *det*, *vol* and *proj* identify a quantity or coordinate convention. The superscripts *E*, *pre*, *ROI*, *meas* and *calc* label an Ewald-selected quantity, a pre-selection quantity, a region-of-interest average, a measurement and a calculation, respectively. These superscripts identify the role of each quantity.

The laboratory axes are y along the nominal incident beam, z vertical and x transverse. The reciprocal basis vectors include 2π ; for the hexagonal setting used here, the basal lattice constant is a , the normal repeat is c , and $|\mathbf{c}^*| = 2\pi/c$. The source-state index is s throughout. The index j follows the main text: it labels an Ewald-intersection branch in Sec. S4 and a parent-rich population in Sec. S7. Parentheses in $S_{hk}^{(j)}$ identify that population. Subscripts are suppressed only where a fixed rod, source state or population is stated explicitly.

Table S1. Coordinates used in the main text and Supporting Information. Reciprocal basis vectors include 2π .

Symbol	Meaning
(h, k, ℓ)	Integer Miller indices; Bragg orders along a rod occur at $L = \ell$.
L	Intrinsic dimensionless coordinate along a crystallite-frame reciprocal rod.
$q_z = \mathbf{c}^* L$	Crystallite-frame film-normal momentum component, as in the main-text stacking model.
Q_z, L_{proj}	Projection along the mean film normal, with $L_{\text{proj}} = Q_z/ \mathbf{c}^* $.
Q_R, r	In-plane momentum magnitude and discrete radial-family label $r = h^2 + hk + k^2$.
α, β	Crystallite tilt magnitude and azimuth in the two-angle orientation map.
$2\theta, \phi_f$	Scattering angle and outgoing-ray azimuth in the declared detector display frame.
$x', z'; x_D, y_D$	Main-text in-plane detector axes and SI column/row axes, with $x_D = x'$ and $y_D = -z'$ in the reference pose.

S1.1 Source state and sample intersection

An incident state s supplies a ray origin $\mathbf{r}_{0,s}$ and an external incident vector

$$\mathbf{k}_{i,s} = \mathbf{k}_i(|\mathbf{k}_{i,s}|, \theta_{i,s}, \phi_{i,s}), \quad |\mathbf{k}_{i,s}| = \frac{2\pi}{\lambda_s}. \quad (\text{S1})$$

Here λ_s is the source wavelength, $\theta_{i,s}$ is the grazing incidence angle and $\phi_{i,s}$ is the incident azimuth in the declared laboratory frame. The subscripts i and f denote incident and exit quantities. The source-position and angular variables are sampled from the measured Gaussian beam distributions, while λ_s is sampled from the specified wavelength spectrum. Writing $\hat{\mathbf{s}}_{i,s} = \mathbf{k}_{i,s}/|\mathbf{k}_{i,s}|$, the incident line, parameterized by path length τ , is

$$\mathbf{r}_s(\tau) = \mathbf{r}_{0,s} + \tau \hat{\mathbf{s}}_{i,s}, \quad \tau \geq 0. \quad (\text{S2})$$

For the following single-ray geometry equations, the source-state subscript s is suppressed. Let the sample entrance plane pass through $\mathbf{r}_S$ and have outward unit normal $\hat{\mathbf{n}}$. Solving $(\mathbf{r} - \mathbf{r}_S) \cdot \hat{\mathbf{n}} = 0$ gives

$$\tau_{\text{hit}} = \frac{(\mathbf{r}_S - \mathbf{r}_0) \cdot \hat{\mathbf{n}}}{\hat{\mathbf{s}}_i \cdot \hat{\mathbf{n}}}, \quad \mathbf{r}_{\text{hit}} = \mathbf{r}_0 + \tau_{\text{hit}} \hat{\mathbf{s}}_i. \quad (\text{S3})$$

The sample-intersection test rejects parallel rays and solutions with $\tau_{\text{hit}} \leq 0$. For orthonormal in-plane sample directions $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_2$, define

$$\xi_1 = (\mathbf{r}_{\text{hit}} - \mathbf{r}_S) \cdot \hat{\mathbf{e}}_1, \quad \xi_2 = (\mathbf{r}_{\text{hit}} - \mathbf{r}_S) \cdot \hat{\mathbf{e}}_2. \quad (\text{S4})$$

A rectangular support with width W along $\hat{\mathbf{e}}_1$ and height H along $\hat{\mathbf{e}}_2$ accepts the state only if

$$|\xi_1| \leq \frac{W}{2}, \quad |\xi_2| \leq \frac{H}{2}. \quad (\text{S5})$$

The illuminated footprint is the weighted distribution of the accepted (ξ_1, ξ_2) values. The source distribution and intersection test determine the footprint directly. For the convention in which $\hat{\mathbf{n}}$ points out of the film, the state-specific grazing angle is

$$\theta_{i,s} = \sin^{-1}(-\hat{\mathbf{s}}_i \cdot \hat{\mathbf{n}}). \quad (\text{S6})$$

The ordered-film configurations used for the displayed Bi_2Se_3 and Bi_2Te_3 calculations specify an unbounded in-plane sample plane. Equation (S5) documents the available finite-support model. The retained unbounded-plane geometry excludes sample-edge clipping from the fitted broadening mechanisms.

S1.2 Detector frame and accepted rays

The detector mapping is most transparent in its own rigid coordinate frame. Its x_D and y_D coordinates increase with detector column and row, respectively, and its plane is $z_D = 0$. In the untilted main-text geometry, x_D follows $+x'$ and y_D follows $-z'$, so $+z_D$ is opposite the outward normal $\hat{\mathbf{n}}_{\text{det}}$. Thus, the main-text z' axis lies in the detector plane, whereas the SI z_D axis is normal to it. Let $\mathbf{R}_D$ and $\mathbf{t}_D$ map detector-frame vectors and points into the laboratory frame. For an exit ray that begins at $\mathbf{r}_{\text{hit}}$ and has unit laboratory direction $\hat{\mathbf{s}}_f$, the detector-frame origin and direction are

$$\mathbf{o}_D = \mathbf{R}_D^T(\mathbf{r}_{\text{hit}} - \mathbf{t}_D), \quad \mathbf{s}_D = \mathbf{R}_D^T \hat{\mathbf{s}}_f. \quad (\text{S7})$$

The detector plane is $z_D = 0$. Its forward intersection is

$$v_D = -\frac{o_{D,z}}{s_{D,z}}, \quad \mathbf{p}_D = \mathbf{o}_D + v_D \mathbf{s}_D, \quad (\text{S8})$$

provided $s_{D,z}$ is nonzero and $v_D > 0$. The front-face convention additionally requires the exit direction to approach the active face. If (x_0, y_0) is the calibrated reference coordinate and p_c, p_r are the column and row pitches, the dimensionless column and row coordinates are

$$c_D = x_0 + \frac{p_{D,x}}{p_c}, \quad r_D = y_0 + \frac{p_{D,y}}{p_r}. \quad (\text{S9})$$

For a detector with N_c columns and N_r rows, the active-area condition is

$$-\frac{1}{2} \leq c_D \leq N_c - \frac{1}{2}, \quad -\frac{1}{2} \leq r_D \leq N_r - \frac{1}{2}. \quad (\text{S10})$$

The subscripts in (c_D, r_D) distinguish detector coordinates from the lattice repeat c and radial-family label r . The beam-center pair (x_0, y_0) follows the main-text calibration convention, in pixel units. The nominal sample-to-detector distance D , detector tilts (β_D, γ_D) , and beam center are contained in $\mathbf{R}_D$, $\mathbf{t}_D$, and the reference coordinate. Here D denotes the separation along the nominal beam axis in the untilted reference pose. A tilted detector is treated through the full rigid transform. Native pixel masses or finite angular-bin quantities are obtained by integrating the continuous detector-coordinate density over the corresponding region. The coordinate Jacobian that defines the density includes the detector solid-angle factor exactly once.

S1.3 Angular display convention

For the angular display, 2θ is measured from the declared incident reference direction. The outgoing-ray azimuth ϕ_f is measured in the plane perpendicular to that direction, from the upward reference toward the negative column direction. In the untilted laboratory geometry this is from $+z$ toward $-x$. Equivalently, for components in that geometry, $\phi_f = \text{atan2}(-s_{f,x}, s_{f,z})$, wrapped to $[-\pi, \pi)$. The same fixed nominal reference is used to remap measured and calculated detector images. This display coordinate is distinct from the incident azimuth ϕ_i and crystallite azimuth β .

S1.4 Source sampling and wavelength dependence

The source representation samples two transverse positions, two propagation angles, and wavelength. Seeded equal-mass antithetic Latin-hypercube states carry weights $w_s = 1/N_s$, where N_s is the source-state count. The position and angular distributions specify the incident beam, whereas the continuous mosaic distribution specifies crystallite orientation. They enter different parts of the calculation. Numerical convergence is discussed in Sec. S8.

For source-state wavelength λ_s and incident direction $\hat{\mathbf{s}}_{i,s}$, define

$$k_s = \frac{2\pi}{\lambda_s}, \quad \mathbf{k}_{i,s} = k_s \hat{\mathbf{s}}_{i,s}. \quad (\text{S11})$$

In reciprocal-space coordinates centered on the origin, the corresponding Ewald sphere is centered at $-\mathbf{k}_{i,s}$ and has radius k_s . The integer Bragg vector is $\mathbf{G}_{hkl} = h\mathbf{a}^* + k\mathbf{b}^* + \ell\mathbf{c}^*$. An allowed reciprocal vector contributes only when

$$|\mathbf{k}_{i,s} + \mathbf{G}_{hkl}| = k_s \quad (\text{S12})$$

within the declared instrumental and numerical resolution.[1] The incident direction is resampled as well when beam divergence is included. Changing wavelength therefore moves the Ewald-sphere center and changes its radius. The calculation reconstructs and tests Eq. (S12) for every wavelength and direction state. Each wavelength therefore requires its own sphere center as well as its own radius.

For one lattice-plane spacing d at fixed orientation and Bragg angle θ_B , Bragg's law gives a useful limiting check,

$$2d \sin \theta_B = \lambda, \quad d\theta_B = \frac{d\lambda}{\lambda} \tan \theta_B. \quad (\text{S13})$$

This scalar derivative provides a local check on the radial direction in which wavelength changes the detector position. The full detector-space condition retains the coupling among sample orientation, incident direction, mosaicity, refraction, and detector acceptance.

S2 Structural intensity along reciprocal rods

The main-text structure factor defines the scattering amplitude of an atomic motif. Finite coherence along the stacking direction converts that amplitude into continuous intensity along a rod. Random in-plane orientation then distributes the rod intensity around a cylinder.

S2.1 Atomic amplitudes and finite stacking

For in-plane indices (h, k) , the unrotated rod is

$$\mathbf{G}_{hk}(L) = h\mathbf{a}^* + k\mathbf{b}^* + L\mathbf{c}^*. \quad (\text{S14})$$

The atomic motif amplitude has the phase convention $\exp(i\mathbf{G} \cdot \mathbf{r}_a)$, where $\mathbf{G}$ is the reciprocal vector and $\mathbf{r}_a$ is the Cartesian position of site a . Atomic form factors include their wavelength-dependent anomalous terms. For a sequence of N layers with motif amplitudes $F_{hk,n}$ and reference-position translations $\mathbf{t}_n$, the coherent amplitude is

$$A_{N,hk}(L, \lambda_s) = \sum_{n=0}^{N-1} F_{hk,n}(L, \lambda_s) \exp[i\mathbf{G}_{hk}(L) \cdot \mathbf{t}_n]. \quad (\text{S15})$$

Here n labels layers from 0 to $N - 1$, and $\mathbf{t}_n$ locates their motif origins in the reference crystallite frame. For fixed (h, k, L, λ_s) , write $A_N = A_{N,hk}(L, \lambda_s)$. The total finite-stack strength is proportional to $|A_N|^2$. An ensemble of stacking sequences instead gives $\langle |A_N|^2 \rangle$, where angle brackets denote the ensemble average. The total-stack and per-layer conventions differ by N and must remain consistent with the population and intensity scale. Section S7 uses an explicitly stated per-trilayer convention.

For identical motifs of amplitude F at one registry, separated by d along the stacking direction, Eq. (S15) reduces to

$$|A_N|^2 = |F|^2 \left| \sum_{n=0}^{N-1} e^{inq_z d} \right|^2 = |F|^2 \frac{\sin^2(Nq_z d/2)}{\sin^2(q_z d/2)}, \quad q_z = |\mathbf{c}^*|L. \quad (\text{S16})$$

The removable limit at $q_z d = 2\pi m$, for integer m , is $N^2|F|^2$. This same-registry example explains finite-width peaks and fringes. A material-specific ordered stack also includes the basal translations of its crystallographic repeat. The native R-centered off-axis Bi_2Se_3 sequence requires the spacing $d = c/3$ together with its basal translations and crystallographic centering phases in Eq. (S15).

S2.2 Displacement conventions

The main text uses Cartesian atomic positions in the phase factor. At an integer Bragg order, the same structure factor can be written in fractional coordinates as

$$F_{hk\ell}(\lambda_s) = \sum_a o_a f_a(Q, \lambda_s) T_a(hk\ell) \exp[2\pi i(hx_a + ky_a + \ell z_a)]. \quad (\text{S17})$$

Here a labels atomic sites, o_a is site occupancy, f_a is the atomic scattering factor including anomalous terms, and (x_a, y_a, z_a) are fractional site coordinates. The magnitude is $Q = |\mathbf{G}_{hk\ell}|$, and $F_{hk\ell}(\lambda_s) = F_{hk}(\ell, \lambda_s)$ uses the main-text rod amplitude at an integer Bragg order. We use

atomic displacement parameter (ADP) for U or B and Debye–Waller factor for the dimensionless amplitude factor T_a . [2] For an isotropic mean-square displacement $U_{\text{iso},a}$ along any one Cartesian direction,

$$T_a(hk\ell) = \exp \left[-8\pi^2 U_{\text{iso},a} \left(\frac{\sin \theta}{\lambda_s} \right)^2 \right] = \exp \left(-\frac{1}{2} U_{\text{iso},a} Q^2 \right), \quad (\text{S18})$$

with $B_{\text{iso},a} = 8\pi^2 U_{\text{iso},a}$ and $Q = 4\pi \sin \theta / \lambda_s$. Here θ is half the scattering angle. Both U and B have units of length squared. The total isotropic mean-square displacement is $3U_{\text{iso},a}$.

The CIF interface accepts isotropic U_{iso} or B_{iso} values and stores their equivalent isotropic mean-square displacement. The shared directional model uses one uniaxial displacement tensor for all sites, expressed in Cartesian coordinates relative to the mean film normal,

$$\mathbf{U}_{\text{film}} = U_R (\mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}}^\top) + U_z \hat{\mathbf{n}}\hat{\mathbf{n}}^\top, \quad (\text{S19})$$

Here $\mathbf{I}$ is the three-dimensional identity matrix. The parameter U_R is the mean-square displacement along either in-plane direction, and U_z is the mean-square displacement along the film normal $\hat{\mathbf{n}}$. The total in-plane value is $2U_R$. Both parameters are nonnegative and have units of length squared. With $Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}$ and $Q_R^2 = |\mathbf{Q}|^2 - Q_z^2$, the resulting Debye–Waller factor is

$$T_{\text{dir}}(\mathbf{Q}) = \exp \left[-\frac{1}{2} (U_R Q_R^2 + U_z Q_z^2) \right]. \quad (\text{S20})$$

This shared factor replaces the site isotropic factors within this parameterization. It multiplies the amplitude, so its intensity factor is $|T_{\text{dir}}|^2$. The tensor is fixed relative to the mean film normal and describes effective directional damping. A displacement tensor tied to an individual crystallite instead rotates with that crystallite, as $\mathbf{U} = \mathbf{R}\mathbf{U}_{\text{cryst}}\mathbf{R}^\top$, where $\mathbf{R}$ is its orientation matrix. General site-specific CIF tensors also require conversion from the stated crystallographic convention to a Cartesian basis. [2]

S2.3 Rod coverage and radial families

Reflection identities are constructed before source integration. If Q_{max} is the largest momentum transfer subtended by the active detector over the wavelength interval and ΔQ is a declared coverage margin, the symbolic candidate set is

$$\mathcal{R} = \left\{ (h, k, \ell) \in \mathbb{Z}^3 : \text{allowed}(h, k, \ell), |\mathbf{G}_{hk\ell}| \leq Q_{\text{max}} + \Delta Q \right\}. \quad (\text{S21})$$

Here *allowed* imposes the reflection conditions of the structural model. For the rod representation, integer (h, k) identities are enumerated by the same detector-coverage rule and the coordinate along each rod remains continuous.

In vacuum, an orientation-independent necessary reach condition is $Q_{R,hk} \leq 2k_s$, with $k_s = 2\pi/\lambda_s$. The shortest sampled wavelength gives the largest such bound. This bound supplies candidates, while the orientation distribution, optical propagation and active detector impose further restrictions. For a hexagonal basal plane, $Q_R = 4\pi\sqrt{r}/(\sqrt{3}a)$ with $r = h^2 + hk + k^2$. Each *allowed* (h, k) member is evaluated individually. The explicit sum accounts for crystallographic multiplicity.

S2.4 Rod measure and powder-cylinder normalization

The main text uses a structural density $S_{hk}(L, \lambda_s)$ per unit L . Let $S_{q,hk}$ denote the corresponding density per dimensional crystallite-frame coordinate q_z . Conservation of intensity requires

$$q_z = |\mathbf{c}^*|L, \quad S_{hk}(L, \lambda_s) dL = S_{q,hk}(q_z, \lambda_s) dq_z, \quad S_{hk} = |\mathbf{c}^*|S_{q,hk}. \quad (\text{S22})$$

Let P_{hk} denote an optional relative population weight of an individual rod. It is separate from the structural density S_{hk} and is included in the combined weight $W_{hk,j,s}$ when the detector density is formed in Sec. S4, following the main-text convention. Uniform azimuth assigns the fraction $d\beta/(2\pi)$ to each azimuth interval. For $Q_R > 0$, the physical cylinder area and its intensity density are

$$dA_{\text{cyl}} = Q_R |\mathbf{c}^*| d\beta dL, \quad \rho_{\text{cyl},hk} = \frac{P_{hk} S_{hk}}{2\pi Q_R |\mathbf{c}^*|} = \frac{P_{hk} S_{q,hk}}{2\pi Q_R}. \quad (\text{S23})$$

Integration around the circumference recovers the original rod strength. The axial rod has $Q_R = 0$ and requires the separate axial treatment because its cylinder circumference vanishes. The full orientation-map determinant in Sec. S4.1 already contains the azimuthal metric factor.

S3 Mosaic orientation and axial scattering

Crystallite tilt changes which parts of a reciprocal rod meet the elastic condition. This section defines the angular probability used for that weighting, then separates transverse orientation sensitivity from the radial axial extension discussed in the main text.

S3.1 Wrapped mosaic distribution

The wrapped mosaic profile uses a signed tilt coordinate $\Delta\theta \in (-\pi, \pi]$ relative to the substrate normal. Its periodic continuation identifies angles differing by 2π . The component normalization below is with respect to $d\Delta\theta$. The nonnegative tilt magnitude and the density per solid angle use different measures, as specified below.

The profile is a wrapped Gaussian G_w plus a wrapped Lorentzian L_w . Their mixture weight is p_{tail} , the Lorentzian half-width is Γ , and the Gaussian standard deviation is σ :

$$\omega_w(\Delta\theta; p_{\text{tail}}, \Gamma, \sigma) = p_{\text{tail}} L_w(\Delta\theta; \Gamma) + (1 - p_{\text{tail}}) G_w(\Delta\theta; \sigma), \quad 0 \leq p_{\text{tail}} \leq 1.$$

The Lorentzian half-width Γ and Gaussian standard deviation σ are independent. The wrapped Lorentzian and wrapped Gaussian are

$$\begin{aligned} L_w(\Delta\theta; \Gamma) &= \sum_{q=-\infty}^{\infty} \frac{1}{\pi} \frac{\Gamma}{(\Delta\theta + 2\pi q)^2 + \Gamma^2} = \frac{1}{2\pi} \frac{\sinh \Gamma}{\cosh \Gamma - \cos \Delta\theta}, \\ G_w(\Delta\theta; \sigma) &= \sum_{q=-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta\theta + 2\pi q)^2}{2\sigma^2}\right] \\ &= \frac{1}{2\pi} \left[1 + 2 \sum_{n=1}^{\infty} e^{-n^2\sigma^2/2} \cos(n\Delta\theta) \right]. \end{aligned}$$

The integer q labels periodic images and n labels positive Fourier harmonics. Angles in these expressions are in radians. Both components integrate to unity over $(-\pi, \pi]$, so ω_w is normalized.

The underlying unwrapped Lorentzian FWHM is 2Γ , and the underlying unwrapped Gaussian FWHM is $2\sqrt{2\ln 2}\sigma$. The local small-width limit recovers the usual line-shape widths, but the refinement and normalization are defined by the wrapped functions above. The broad functional form is phenomenological. These equations define the two-component model. Establishing its necessity requires comparison with a re-optimized Gaussian-only model.

S3.2 Folded tilt probability and orientation measure

For the even signed profile above, the probability density of the magnitude $0 \leq \alpha \leq \pi$ is

$$p_\alpha(\alpha) = \omega_w(\alpha) + \omega_w(-\alpha) = 2\omega_w(\alpha), \quad p_m(\alpha, \beta) = \frac{p_\alpha(\alpha)}{2\pi}. \quad (\text{S24})$$

Thus $\int_0^\pi p_\alpha d\alpha = 1$ and $\int_0^\pi \int_0^{2\pi} p_m d\beta d\alpha = 1$. The product $p_m d\alpha d\beta$ is the probability of a crystallite orientation within that angular interval. When the same normal directions are expressed as a density per solid angle, $d\Omega = \sin \alpha d\alpha d\beta$, the corresponding density is

$$f_\Omega(\alpha, \beta) = \frac{p_\alpha(\alpha)}{2\pi \sin \alpha}, \quad 0 < \alpha < \pi. \quad (\text{S25})$$

The apparent polar singularity is integrable with the solid-angle measure. On a sphere of radius Q , the density per surface area is f_Ω/Q^2 . A smooth density specified directly per solid angle is a different orientation law. Figure S1 shows these probability conventions.

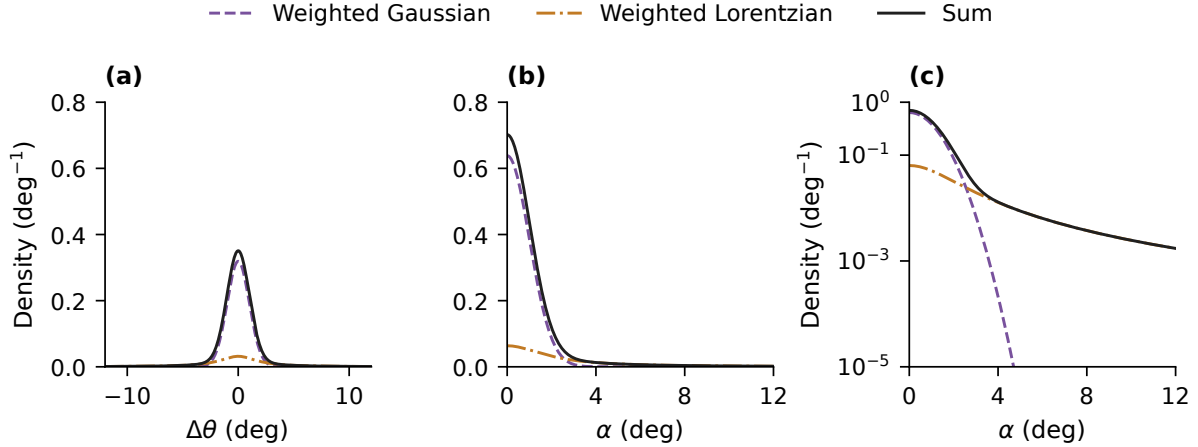


Figure S1. Wrapped angular distributions and their measures. Purple dashed and amber dash-dotted lines show the weighted Gaussian and Lorentzian components, and the black solid line is their sum. (a) Signed density versus $\Delta\theta$. (b) The folded magnitude density on the same linear ordinate scale. (c) The same folded curves on a logarithmic ordinate scale. The parameter values are $\sigma = 1^\circ$, $\Gamma = 2^\circ$, and $p_{\text{tail}} = 0.2$, with azimuth uniform. Ordinates are probability density per degree. Each component is normalized over the complete angular domain before applying its mixture weight, and the displayed windows retain that full-domain normalization. Folding adds the positive and negative signed contributions. The wider component remains visible away from the core.

The orientation map rotates the entire reciprocal rod, including its basal offset. For the positive- L axial rod, the geometry simplifies to

$$Q_R = |\mathbf{c}^*|L \sin \alpha, \quad Q_z = |\mathbf{c}^*|L \cos \alpha. \quad (\text{S26})$$

Consequently, even an axial point has $L_{\text{proj}} = L \cos \alpha$ after tilt. For an off-axis rod, the rotated basal offset also contributes to projected Q_z . Figure S2 reserves the material-specific component comparison.

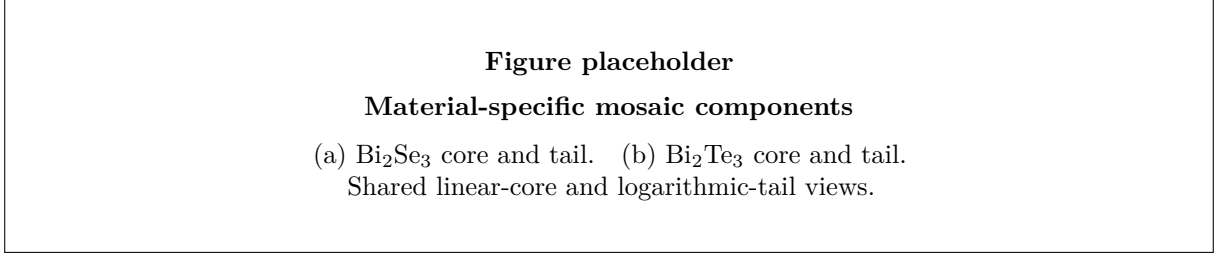


Figure S2. Placeholder for material-specific weighted mosaic components. Direct curves and their matching parameter state are to be inserted with the declared probability measure, full-support normalization, and parameter uncertainties where available.

S3.3 Controls on the radial axial extension

The contribution of wavelength spread to an axial feature can be examined with a mean-wavelength control at fixed remaining inputs. The full detector quantity is

$$I_{\text{det}}^{\text{full}}(c_D, r_D) = \sum_s w_s I_{\text{det},s}(c_D, r_D; \lambda_s), \quad \sum_s w_s = 1, \quad (\text{S27})$$

Here $I_{\text{det},s}(c_D, r_D; \lambda)$ denotes the detector-coordinate intensity density for the position and direction of state s , evaluated at the indicated wavelength λ . Every state has its own incident wavevector and Ewald intersection. The superscripts full and mono identify the full-spectrum and mean-wavelength calculations. The control replaces the distribution by its weighted mean,

$$I_{\text{det}}^{\text{mono}}(c_D, r_D) = \sum_s w_s I_{\text{det},s}(c_D, r_D; \bar{\lambda}), \quad \bar{\lambda} = \sum_s w_s \lambda_s, \quad (\text{S28})$$

while holding source positions and directions, their weights, geometry, structural and mosaic parameters, intensity scale, background treatment, and detector mapping fixed. Wavelength-dependent scattering factors and optical quantities are evaluated consistently at the control wavelength.

In the Bi₂Se₃ and Bi₂Te₃ model controls, the radial axial extension persisted when wavelength and incident-direction spread were removed. It also persisted when the source was reduced to one monochromatic point ray at the configured mean position and direction. These controls show that the tested model generates the extension even for a monochromatic source.

The model produces continuous finite-stack axial intensity weighted by the orientation distribution near the mean specular direction. In the Bi₂Se₃ control, changing incidence moved the radial extension from the vicinity of 003 to 006 while preserving the structural Bragg positions. Increasing the coherent repeat count reduced the far-tail intensity relative to the integrated Bragg strength. These interventions identify the origin of the extension within the tested model state.

The cause of the measured feature and the relative-order intensity disagreement remain unresolved by these calculations. Radial and transverse profiles provide separate tests of source and orientation effects. Assessing a wavelength effect requires a matched control for the stated geometry and model. Figures S3 and S4 reserve the direct control calculations and the separate test of component necessity.

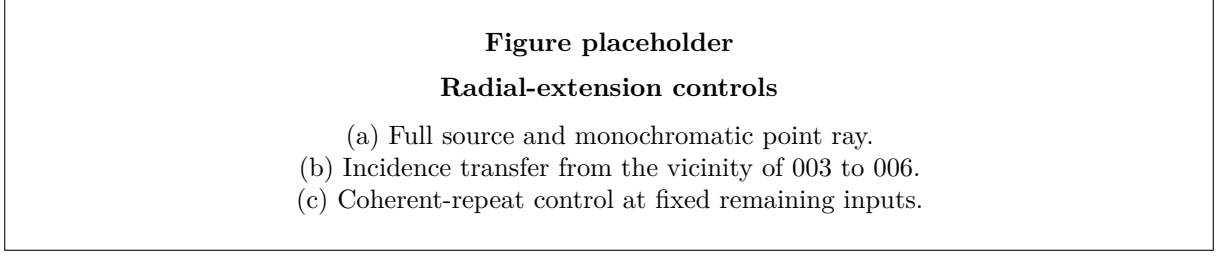


Figure S3. Placeholder for direct model controls on the radial axial extension. The intended panels compare identical detector regions and declared common scales while varying source spread, incidence, or coherent repeat count separately. The controls test the retained model mechanism.

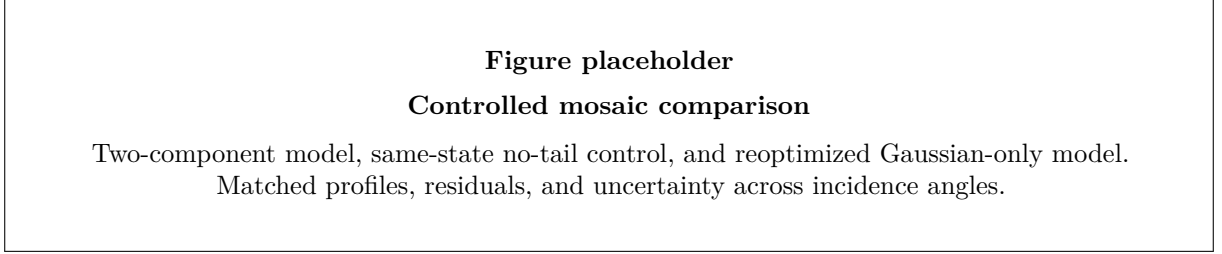


Figure S4. Placeholder for the controlled test of a broad mosaic component. A comparison requires direct calculated arrays, identical measured regions, declared nuisance parameters, and a common residual criterion.

S4 Ewald selection and detector profiles

The Ewald condition selects points from the oriented reciprocal rods. Their intensity is transformed to detector coordinates before a finite integration region defines a measured or calculated profile. This is the technical counterpart of the main-text Ewald-to-detector construction.

S4.1 Forward Ewald-surface chart and sequential Jacobians

The local Ewald chart below connects the structural density per dL to density per Ewald-surface area. The detector Jacobian then expresses that density per detector-coordinate area.

For one rod, source state, and crystallite orientation, the rotated reciprocal rod is linear in the intrinsic coordinate L ,

$$\mathbf{Q}_{hk}(\alpha, \beta, L) = \mathbf{q}_{hk}(\alpha, \beta) + L \mathbf{d}_{hk}(\alpha, \beta), \quad \mathbf{q}_{hk} = \mathbf{Q}_{hk}(\alpha, \beta, 0), \quad \mathbf{d}_{hk} = \frac{\partial \mathbf{Q}_{hk}}{\partial L}. \quad (\text{S29})$$

Here $\mathbf{q}_{hk}$ is the rotated basal offset and $\mathbf{d}_{hk}$ is the rod tangent per unit L . Let $\mathbf{k}'_{i,s}$ be the real internal phase wavevector and $K_s = |\mathbf{k}'_{i,s}|$. Squaring the elastic condition gives

$$\tilde{g}_{hk,s}(\alpha, \beta, L) = \left| \mathbf{k}'_{i,s} + \mathbf{q}_{hk} + L \mathbf{d}_{hk} \right|^2 - K_s^2 = A_E L^2 + B_E L + C_E, \quad (\text{S30})$$

with

$$A_E = \mathbf{d}_{hk} \cdot \mathbf{d}_{hk}, \quad B_E = 2 \mathbf{d}_{hk} \cdot (\mathbf{k}'_{i,s} + \mathbf{q}_{hk}), \quad C_E = \left| \mathbf{k}'_{i,s} + \mathbf{q}_{hk} \right|^2 - K_s^2. \quad (\text{S31})$$

For discriminant $\Delta_E = B_E^2 - 4A_EC_E > 0$, the regular Ewald branches are

$$L_{hk,j,s}(\alpha, \beta) = \frac{-B_E \pm \sqrt{\Delta_E}}{2A_E}, \quad \mathbf{Q}_{hk,j,s}^E(\alpha, \beta) = \mathbf{Q}_{hk}[\alpha, \beta, L_{hk,j,s}(\alpha, \beta)]. \quad (\text{S32})$$

A negative discriminant means that the oriented rod misses the Ewald sphere, zero gives tangency, and a positive discriminant gives two algebraic branches. At tangency, the regular-root Jacobian is singular and the finite-region limit needs separate treatment. Equation (S32) makes (α, β) the independent coordinates of a local two-dimensional Ewald-surface patch, while L is the dependent coordinate selected by elasticity.

For derivatives of the surface chart it is convenient to use the unsquared residual

$$g_{hk,s}(\alpha, \beta, L) = \left| \mathbf{k}'_{i,s} + \mathbf{Q}_{hk}(\alpha, \beta, L) \right| - K_s. \quad (\text{S33})$$

For a fixed rod and source state, write $L_j = L_{hk,j,s}$ and suppress the same indices on $\mathbf{Q}^E$ and J_E . On a regular branch, implicit differentiation gives

$$\frac{\partial L_j}{\partial \mu} = -\frac{\partial_\mu g_{hk,s}}{\partial_L g_{hk,s}}, \quad \frac{\partial \mathbf{Q}^E}{\partial \mu} = \frac{\partial \mathbf{Q}}{\partial \mu} + \frac{\partial \mathbf{Q}}{\partial L} \frac{\partial L_j}{\partial \mu}, \quad \mu \in \{\alpha, \beta\}. \quad (\text{S34})$$

The physical surface element is therefore

$$dA_Q = \left| \frac{\partial \mathbf{Q}^E}{\partial \alpha} \times \frac{\partial \mathbf{Q}^E}{\partial \beta} \right| d\alpha d\beta = J_E J_{\text{vol},hk} d\alpha d\beta, \quad (\text{S35})$$

with the real internal exit vector $\mathbf{k}'_{f,s} = \mathbf{k}'_{i,s} + \mathbf{Q}^E$ and Jacobians

$$J_{\text{vol},hk} = |\det[\partial_\alpha \mathbf{Q}_{hk}, \partial_\beta \mathbf{Q}_{hk}, \partial_L \mathbf{Q}_{hk}]|, \quad J_E = \frac{1}{|\partial_L g_{hk,s}|} = \frac{K_s}{|\mathbf{k}'_{f,s} \cdot \mathbf{d}_{hk}|}. \quad (\text{S36})$$

The optical imaginary wavevector component controls attenuation, while the real phase vectors define the Ewald sphere. The measure formulation keeps the four objects used in the main text separate: $S_{hk}(L, \lambda_s)$ is the structural rod intensity, $\mathbf{Q}_{hk}(\alpha, \beta, L)$ is the location map, $p_m(\alpha, \beta)$ is the orientation weight, and $J_{\text{vol},hk}$ is the geometric change of measure. The pre-Ewald intensity measure is

$$dI_{hk,s}^{\text{pre}} = p_m(\alpha, \beta) S_{hk}(L, \lambda_s) d\alpha d\beta dL. \quad (\text{S37})$$

Restricting this measure to the regular roots gives

$$dI_{hk,j,s}^E = p_m(\alpha, \beta) S_{hk}(L_j, \lambda_s) J_E d\alpha d\beta. \quad (\text{S38})$$

Dividing Eq. (S38) by Eq. (S35) cancels the line-sphere factor. Including the combined weight $W_{hk,j,s}$ then gives the physical Ewald-surface density

$$\rho_{Q,s}^E(\mathbf{Q}) = \sum_{\substack{(h,k),j,\alpha,\beta \\ \mathbf{Q}_{hk,j,s}^E(\alpha,\beta)=\mathbf{Q}}} W_{hk,j,s} \frac{p_m(\alpha, \beta) S_{hk}(L_j, \lambda_s)}{J_{\text{vol},hk}}. \quad (\text{S39})$$

Here $W_{hk,j,s}$ includes the individual-rod population P_{hk} and the source-dependent optical factors; each factor enters once. The sum includes all compatible rods, root branches and orientations.

The source weight w_s is applied separately when the detector images are accumulated. All branch-dependent factors are evaluated at $L = L_{hk,j,s}$. Thus J_E enters the intermediate restricted measure and cancels when that measure is divided by the Ewald-surface area. The reciprocal-volume factor remains because it converts the original (α, β, L) measure into physical reciprocal-space density.

The notation $J_{\text{det},s}$ is the detector Jacobian used in the main text. For continuous detector coordinates (c_D, r_D) , let $\mathbf{Q}_s(c_D, r_D)$ be the internal momentum transfer selected by that detector point and source state. Define

$$J_{\text{det},s} = \left| \frac{\partial \mathbf{Q}_s}{\partial c_D} \times \frac{\partial \mathbf{Q}_s}{\partial r_D} \right|, \quad dA_Q = J_{\text{det},s} dc_D dr_D. \quad (\text{S40})$$

The source-resolved detector-coordinate density is then

$$I_{\text{det},s}(c_D, r_D) = \sum_{\text{compatible}} W_{hk,j,s} p_m(\alpha, \beta) S_{hk}(L, \lambda_s) \frac{J_{\text{det},s}}{J_{\text{vol},hk}}, \quad (\text{S41})$$

and the final detector density and mass I_p integrated over detector pixel p are

$$I_{\text{det}}(c_D, r_D) = \sum_s w_s I_{\text{det},s}(c_D, r_D), \quad I_p = \int_{\text{pixel } p} I_{\text{det}}(c_D, r_D) dc_D dr_D. \quad (\text{S42})$$

Equation (S41) shows the sequential change of measure directly: the reciprocal-volume Jacobian enters first and the detector-coordinate Jacobian enters second, leaving the net geometric factor $J_{\text{det},s}/J_{\text{vol},hk}$. The detector factor contains the pixel solid angle, detector pose and distance, source-point parallax, and the area change associated with exit refraction. An additional independent solid-angle correction would count this geometric factor twice.

Finally, the powder-cylinder factor $1/(2\pi Q_R)$ and the full determinant $J_{\text{vol},hk}$ are two stages of the same azimuthal change of measure. The former expresses the density per cylinder area after distributing a rod uniformly in β . In the direct pushforward from $d\alpha d\beta dL$ to d^3Q , the same β -arc-length geometry is already contained in $J_{\text{vol},hk}$. The determinant therefore accounts for this circumference contribution exactly once.

S4.2 Axial root and detector-coordinate evaluation

For a fixed source state, suppress s on the incident vector. The axial rod passes through the reciprocal origin. For its rotated unit direction $\hat{\mathbf{d}}$, write $\mathbf{Q}(q_z) = q_z \hat{\mathbf{d}}$. The elastic condition becomes

$$|\mathbf{k}'_i + q_z \hat{\mathbf{d}}|^2 = |\mathbf{k}'_i|^2, \quad q_z(2\mathbf{k}'_i \cdot \hat{\mathbf{d}} + q_z) = 0. \quad (\text{S43})$$

The root $q_z = 0$ is the direct-beam point. The nonzero kinematic root is $q_z = -2\mathbf{k}'_i \cdot \hat{\mathbf{d}}$, with $L = q_z/|\mathbf{c}^*|$. For a nonunit direction $\mathbf{d}$ and coordinate v in $\mathbf{Q} = v\mathbf{d}$, the second root is $v = -2\mathbf{k}'_i \cdot \mathbf{d}/(\mathbf{d} \cdot \mathbf{d})$.

The forward roots provide a geometric construction. In detector-coordinate evaluation, a detector point fixes the outgoing air ray, exit refraction fixes the internal phase vector, and their difference supplies $\mathbf{Q}_s$. The inverse reciprocal map then returns all compatible rods, orientations and intrinsic L values. Both descriptions use the density in Eq. (S41). Since $dA_Q = K_s^2 d\Omega$, density per internal outgoing solid angle is $K_s^2 \rho_{Q,s}^E$.

S4.3 Finite integration and projected coordinates

Here $d\Omega$ is an element of internal outgoing solid angle. The detector image remains the primary data object. The 2θ - ϕ_f representation is an overlap-weighted remapping used to evaluate fixed- r or fixed- Q_R trajectory masks and projected Q_z or L_{proj} profiles. Its statistical information comes from the original detector image. The same remapping, masks, and profile extraction are applied to measured and calculated detector images.

The caked image is constructed with a sparse detector-to-cake operator

$$\mathbf{M}_{\text{cake}} \in \mathbb{R}^{N_{\text{bin}} \times N_{\text{pix}}}, \quad (\text{S44})$$

where N_{pix} is the number of detector pixels, N_{bin} is the number of caked bins, and M_{bp}^{cake} is the fractional overlap weight from detector pixel p into caked bin b . For detector signal s_p and normalization n_p , define accumulated caked-bin signal S_b , normalization N_b and normalized intensity I_b :

$$S_b = \sum_p M_{bp}^{\text{cake}} s_p, \quad N_b = \sum_p M_{bp}^{\text{cake}} n_p, \quad I_b = \frac{S_b}{N_b}, \quad (\text{S45})$$

for bins with $N_b > 0$. Signal and normalization are therefore accumulated before division. This prevents a projected profile from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

For each remapped bin, the calibrated detector and sample geometry define a scattering vector $\mathbf{Q}$. The coordinates used to label selected trajectories are

$$\mathbf{Q}_{\parallel} = \mathbf{Q} - (\mathbf{Q} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}, \quad Q_R = |\mathbf{Q}_{\parallel}|, \quad Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}, \quad L_{\text{proj}} = \frac{Q_z}{|\mathbf{c}^*|} = \frac{Q_z c}{2\pi}, \quad (\text{S46})$$

where $\hat{\mathbf{n}}$ is the film normal and $\mathbf{Q}_{\parallel}$ is the component within its plane. For a hexagonal basal plane, the dimensionless radial-family index associated with (h, k) is $r \equiv h^2 + hk + k^2$. The index r labels a discrete family. Its radius is $Q_R(r)$, and the explicit sum over individual (h, k) members accounts for multiplicity. For target radius $Q_{R,0}$, radial half-width ΔQ_R and accepted projected limits $Q_{z,\text{min}}$ and $Q_{z,\text{max}}$, a fixed- Q_R or fixed- r family is retained when

$$|Q_R - Q_{R,0}| \leq \Delta Q_R, \quad Q_{z,\text{min}} \leq Q_z \leq Q_{z,\text{max}}. \quad (\text{S47})$$

This region is simple in the physical Q_R - Q_z description but appears curved in the displayed 2θ - ϕ_f map because the coordinate transformation is nonlinear. The coordinate projection fully accounts for the curved shape.

For a selected profile-bin index ν , let $\mu_{\nu b}$ be the mask or interpolation weight that assigns caked bin b to that profile point. Write $S_{\nu} = \sum_b \mu_{\nu b} S_b$ and $N_{\nu} = \sum_b \mu_{\nu b} N_b$ for its accumulated signal and normalization. The profile is

$$I_{\nu}^{\text{ROI}} = \frac{S_{\nu}}{N_{\nu}} = \frac{\sum_b \mu_{\nu b} S_b}{\sum_b \mu_{\nu b} N_b} = \frac{\sum_p a_{\nu p} s_p}{\sum_p a_{\nu p} n_p}, \quad a_{\nu p} = \sum_b \mu_{\nu b} M_{bp}^{\text{cake}}. \quad (\text{S48})$$

Each point in the plotted profile is therefore an average over a finite detector-space band. Its coordinate may be labeled by Q_z or L_{proj} , and the point represents a finite-bin projection over the accepted reciprocal-space interval.

Intrinsic L belongs to the crystallite-frame rod. The nominal detector projection L_{proj} uses the mean film normal and the declared reference beam. A fixed detector point generally has a

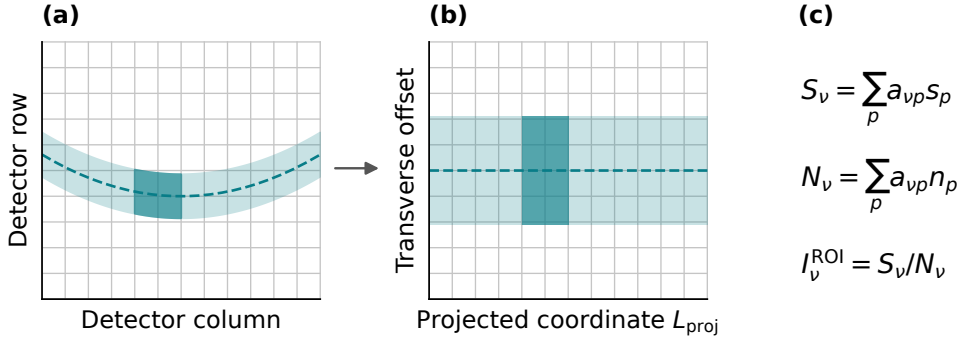


Figure S5. Construction of a finite-region detector profile. (a) A curved integration band crosses native detector pixels. Teal shading marks accepted area, and the darker segment marks one selected profile interval. (b) A coordinate transformation straightens the band into a strip, where the same darker interval is retained. Gray grid lines show pixel or bin boundaries, and the dashed line marks the trajectory center. (c) Detector signal and normalization pass through the same overlap and mask weights before their accumulated values are divided. The equations define an average over the accepted area.

different source-resolved $\mathbf{Q}_s$ for each incident state. Source-resolved physical reciprocal maps and nominal measured-image displays are therefore distinguished. The same nominal display map is applied to measured and calculated detector images.

Coordinate and intensity uncertainty are derived in Sec. S8.3. The finite integration in Fig. S5 is common to the ordered comparisons in Sec. S6 and the selected PbI_2 profiles in Sec. S7.

S5 Refraction, absorption, and reflectivity

The main-text reflectivity discussion separates the low-angle optical response from finite-stack diffraction. The equations below specify the propagation factors and the practical crossover used in the physical diagnostic.

S5.1 Wave propagation, refraction, and absorption

Let $\zeta \geq 0$ denote normal depth into the film and use the field convention $\exp[i(\tilde{\mathbf{k}} \cdot \mathbf{r} - \omega_{\text{em}} t_{\text{time}})]$. For wavelength sample λ_s with source weight w_s ,

$$k_s = \frac{2\pi}{\lambda_s}, \quad \tilde{n}_q(\lambda_s) = 1 - \delta_q(\lambda_s) + i\beta_q(\lambda_s), \quad (\text{S49})$$

Here $\tilde{\mathbf{k}}$ is the complex optical wavevector, $\mathbf{r}$ is position and $\tilde{n}_q$ is the complex refractive index. The prime on an optical wavevector identifies propagation inside the film, as in the main text. The index s is the source-state index, q identifies the optical layer, ω_{em} is angular frequency, and t_{time} is time. The real internal phase vectors used in the main text and Sec. S4 are $\mathbf{k}'_{i,s} = \text{Re} \tilde{\mathbf{k}}'_{i,s}$ and $\mathbf{k}'_{f,s} = \text{Re} \tilde{\mathbf{k}}'_{f,s}$. The symbol $\tilde{k}_\perp$ below denotes the complex component along the local inward normal. The laboratory z axis is defined separately in Sec. S1. The decrement δ_q controls refraction and critical-angle behavior, whereas β_q controls absorption and penetration depth.

Both are wavelength-dependent material inputs. Their source, interpolation, and association with the sampled spectrum are recorded with the calculation configuration.

Tangential matching and the complex normal component.

Translational invariance along a flat interface conserves the tangential wavevector. For a field entering layer q from the ambient medium (index 0), let $\mathbf{k}_{\parallel,q}$ denote the conserved tangential component. The superscript $(+)$ selects the inward, decaying normal root:

$$\mathbf{k}_{\parallel,q}(\lambda_s) = \mathbf{k}_{\parallel,0}(\lambda_s), \quad \tilde{k}_{\perp,q}^{(+)}(\lambda_s) = \sqrt{[\tilde{n}_q(\lambda_s)k_s]^2 - |\mathbf{k}_{\parallel,0}(\lambda_s)|^2}. \quad (\text{S50})$$

The root is chosen so that the field decays in its propagation direction. For the inward solution and the depth convention above, $\text{Im} \tilde{k}_{\perp,q}^{(+)} \geq 0$. [3] The reciprocal exit field is obtained from the same boundary condition with its own conserved tangential component and propagation direction. Real transmitted angles may be reported after this complex-vector calculation. Their evaluation retains both the real and imaginary parts of $\tilde{n}_q$.

Interface amplitudes and normal flux.

For a scalar field at an interface from medium p to medium q , the field transmission amplitude is

$$t_{pq} = \frac{2\tilde{k}_{\perp,p}}{\tilde{k}_{\perp,p} + \tilde{k}_{\perp,q}}. \quad (\text{S51})$$

Equation (S51) specifies the field-amplitude coefficient. Converting it to a transmitted-flux fraction requires the appropriate flux normalization. In the nonabsorbing scalar limit, the associated normal-flux ratio includes the factor $\text{Re} \tilde{k}_{\perp,q} / \text{Re} \tilde{k}_{\perp,p}$ in addition to $|t_{pq}|^2$. Absorbing or polarization-resolved layers require the corresponding Poynting-flux or Fresnel normalization. The calculation must therefore declare whether each interface weight is a field amplitude, an intensity, or a normal flux. The transmitted-flux fraction combines $|t_{pq}|^2$ with that normalization.

Normal-depth attenuation.

Let $\kappa_i(\lambda_s)$ and $\kappa_f(\lambda_s)$ be the nonnegative imaginary normal components for propagation from the surface to a scattering depth and from that depth back to the surface, respectively:

$$\kappa_i(\lambda_s) = \text{Im} \tilde{k}_{\perp,i}^{(+)}(\lambda_s), \quad \kappa_f(\lambda_s) = \text{Im} \tilde{k}_{\perp,f}^{(+)}(\lambda_s), \quad \kappa_i, \kappa_f \geq 0. \quad (\text{S52})$$

Here $\tilde{k}_{\perp,f}^{(+)}$ is the inward reciprocal exit-channel root used by reciprocity. Thus κ_f is the positive attenuation constant for the physical path from depth back to the surface. Equivalently, $\kappa_f = -\text{Im} \tilde{k}_{\perp,f}^{(\text{out})}$ if an outward wavevector is stored explicitly. For a normalized scattering-depth distribution $\rho_{\text{depth}}(\zeta)$ over a film of thickness t , the propagation weight is

$$W_{\text{depth}}(\lambda_s) = \int_0^t \rho_{\text{depth}}(\zeta) \exp[-2(\kappa_i(\lambda_s) + \kappa_f(\lambda_s))\zeta] d\zeta, \quad \int_0^t \rho_{\text{depth}}(\zeta) d\zeta = 1. \quad (\text{S53})$$

For uniform scattering density, this becomes

$$W_{\text{depth}}(\lambda_s) = \frac{1}{t} \int_0^t \exp[-2(\kappa_i(\lambda_s) + \kappa_f(\lambda_s))\zeta] d\zeta \quad (\text{S54})$$

$$= \frac{1 - \exp[-2(\kappa_i(\lambda_s) + \kappa_f(\lambda_s))t]}{2(\kappa_i(\lambda_s) + \kappa_f(\lambda_s))t}. \quad (\text{S55})$$

Equations (S53) and (S55) use normal depth. Multiplying $\text{Im } \tilde{k}_\perp$ by an oblique ray-path length would count the angular projection a second time. The depth weight is combined with consistently normalized entrance and exit interface factors before it multiplies the structure-factor and mosaic weights.

Detector accumulation over the bandwidth.

For each accepted source-state and rod contribution, the wavelength-resolved detector density is formed with the same geometry, structure-factor, mosaic, detector, and optical interfaces. The detector images are then added incoherently,

$$I_{\text{det}}(c_D, r_D) = \sum_s w_s I_{\text{det},s}[c_D, r_D; \lambda_s, \{\tilde{n}_q(\lambda_s)\}]. \quad (\text{S56})$$

Changing λ_s changes the moving Ewald construction, the layer optical constants, the complex normal components, and the interface and depth weights. The planar-interface correction is retained for every wavelength state. Its role is expected to be strongest when an incident or exit angle approaches the critical-angle regime. This optical treatment supplies a propagation correction. Assessing experimental intensity agreement requires measured/calculated comparisons.

S5.2 Parratt-to-kinematic crossover for the axial profile

The detector-derived $r = 0$ profile spans two limiting regimes. The low- Q_{loc} response is described by standard Parratt reflectivity, while the higher- Q_{loc} Bragg response is described by a finite-stack kinematic term. The two branches are matched smoothly over a declared overlap interval. We refer to this matching as the Parratt-to-kinematic crossover. It is a practical connection between two established limiting calculations. The main-text reflectivity figure shows measured Bi_2Se_3 intensities versus detector 2θ .

For the physical crossover diagnostic, the dimensionless coordinate is

$$x = \frac{Q_{\text{loc}}}{Q_c}, \quad L_{\text{loc}} = \frac{Q_{\text{loc}}c}{2\pi} = x \frac{Q_c c}{2\pi}. \quad (\text{S57})$$

Here Q_c is the film critical momentum transfer for total external reflection at wavelength λ_s , and c is the out-of-plane lattice constant. The transfer $Q_{\text{loc}} = |\mathbf{k}_{f,s} - \mathbf{k}_{i,s}|$ is the local momentum-transfer magnitude for specular reflection from a crystallite, whose normal is parallel to that transfer. It generally differs from its mean-film projection Q_z . At nominal incidence θ_i , the mean-film specular direction is $2\theta = 2\theta_i$. Tilted crystallites can contribute axial scattering away from that direction. The L_{loc} value in Eq. (S57) is an external local optical coordinate. Finite-stack scattering uses the separate intrinsic phase coordinate. The diagnostic crossover coordinate uses Q_{loc}/Q_c and the corresponding L_{loc} value defined above.

The low- Q_{loc} branch is the Parratt reflectivity.[4] For a layer q with complex refractive index $\tilde{n}_q = 1 - \delta_q + i\beta_q$, the conserved in-plane wavevector fixes the layer-normal component

$$\tilde{k}_{\perp,q} = \sqrt{(\tilde{n}_q k_s)^2 - |\mathbf{k}_{\parallel}|^2}, \quad k_s = \frac{2\pi}{\lambda_s}. \quad (\text{S58})$$

The fixed source-state label s is suppressed on the layer amplitudes below. For the depth and field convention used above, the square-root branch is chosen to give decay in the relevant propagation direction. The superscript (0) labels the smooth-interface coefficient. The Fresnel amplitude at the interface between layers q and $q+1$ is

$$r_q^{(0)} = \frac{\tilde{k}_{\perp,q} - \tilde{k}_{\perp,q+1}}{\tilde{k}_{\perp,q} + \tilde{k}_{\perp,q+1}}. \quad (\text{S59})$$

If the lower interface has RMS roughness $\sigma_{\text{rough},q}$, this coefficient is damped by

$$r_q = r_q^{(0)} \exp\left[-2\tilde{k}_{\perp,q}\tilde{k}_{\perp,q+1}\sigma_{\text{rough},q}^2\right]. \quad (\text{S60})$$

For a finite layer of thickness d_{q+1} below the interface, the round-trip phase factor $P_{\text{opt},q+1}$ is

$$P_{\text{opt},q+1} = \exp\left(2i\tilde{k}_{\perp,q+1}d_{q+1}\right). \quad (\text{S61})$$

Starting from the semi-infinite substrate and recursing upward, the effective reflection amplitude R_q^{amp} at interface q is

$$R_q^{\text{amp}} = \frac{r_q + R_{q+1}^{\text{amp}}P_{\text{opt},q+1}}{1 + r_q R_{q+1}^{\text{amp}}P_{\text{opt},q+1}}. \quad (\text{S62})$$

The optical branch in this diagnostic is

$$R_{\text{P}}(Q_{\text{loc}}) = |R_0^{\text{amp}}(Q_{\text{loc}})|^2. \quad (\text{S63})$$

This branch retains total external reflection, refraction, evanescence, finite penetration depth, roughness damping, and multiple reflection between the film interfaces.

The high- Q_{loc} branch is built from the normalized finite-stack kinematic term

$$\hat{S}_{00}(L) = \frac{S_{q,00}(|\mathbf{c}^*|L)}{S_{q,00}(0)}. \quad (\text{S64})$$

The hat denotes normalization to the zero-transfer strength, so $\hat{S}_{00}(0) = 1$. The fixed wavelength λ_s is suppressed in these axial structure terms. Here $S_{q,00}$ is the ordered axial structural strength per dimensional rod coordinate from Sec. S2.4. Near the critical edge, the structure term uses the internal film phase. To match the Parratt film phase, the finite-stack coordinate is computed from the real propagating part of the film wavevector,

$$Q_{\text{film}} = \text{Re}\left(2\tilde{k}_{\perp,\text{film}}\right), \quad L_{\text{film}} = \frac{Q_{\text{film}}c}{2\pi}. \quad (\text{S65})$$

Below the critical edge this real part may approach zero in the weak-absorption limit, while a complex internal wavevector still describes evanescence and absorption. The superscript (0) on the kinematic branch below labels its unscaled value:

$$R_{\text{kin}}^{(0)}(Q_{\text{loc}}) = \frac{\hat{S}_{00}(L_{\text{film}})}{Q_{\text{loc}}^2}. \quad (\text{S66})$$

Since $\hat{S}_{00}$ is dimensionless, $R_{\text{kin}}^{(0)}$ has units of length squared. It is an unscaled intensity shape. The Q_{loc}^{-2} factor gives the high- Q_{loc} structural envelope, while the internal coordinate L_{film} keeps the finite-stack fringe phase consistent with the optical film phase.

A multiplicative scale places this kinematic branch on the Parratt intensity scale. The scale is fit in a high- Q_{loc} overlap window, excluding the selected $00L$ neighborhoods used in the diagnostic. With fit-point index i and retained fit set $\mathcal{F}$, the log-median estimate is

$$A_{\text{opt}} = Q_c^2 \exp \left[\text{median}_{i \in \mathcal{F}} \ln \left(\frac{R_P(Q_{\text{loc},i})}{Q_c^2 R_{\text{kin}}^{(0)}(Q_{\text{loc},i})} \right) \right], \quad (\text{S67})$$

with the default overlap window $5 < Q_{\text{loc}}/Q_c < 10$. The factors of Q_c^2 make the logarithm's argument dimensionless and give A_{opt} units of inverse length squared. The scaled kinematic branch is therefore dimensionless, like R_P :

$$R_{\text{kin}}(Q_{\text{loc}}) = A_{\text{opt}} R_{\text{kin}}^{(0)}(Q_{\text{loc}}). \quad (\text{S68})$$

The crossover window is selected by comparing the two positive finite branches in log intensity. Both branches are evaluated at the same dimensional transfer $Q_{\text{loc}} = Q_c x$:

$$\delta_{\log}(x) = \log_{10} \left[\frac{R_{\text{kin}}(Q_c x)}{R_P(Q_c x)} \right]. \quad (\text{S69})$$

Candidate points must lie in the search range $3 \leq x \leq 10$ and satisfy $|\delta_{\log}(x)| \leq 0.10$. The selected window $[x_1, x_2]$ is the widest contiguous eligible region with width at least 1.0 in Q_{loc}/Q_c . Ties are broken by the smaller median $|\delta_{\log}|$. If the search rejects every candidate region, the fallback window is $x_1 = 3$ and $x_2 = 6$.

Inside the selected window, the interpolation coordinate and weight are

$$\eta_{\text{blend}} = \frac{x - x_1}{x_2 - x_1}, \quad w_{\text{blend}}(\eta_{\text{blend}}) = 6\eta_{\text{blend}}^5 - 15\eta_{\text{blend}}^4 + 10\eta_{\text{blend}}^3, \quad (\text{S70})$$

with η_{blend} clipped to the interval $[0, 1]$. The final branch is blended in log reflectivity,

$$R_{\text{blend}}(x) = \begin{cases} R_P(Q_c x), & x \leq x_1, \\ 10^{(1-w_{\text{blend}}) \log_{10} R_P(Q_c x) + w_{\text{blend}} \log_{10} R_{\text{kin}}(Q_c x)}, & x_1 < x < x_2, \\ R_{\text{kin}}(Q_c x), & x \geq x_2. \end{cases} \quad (\text{S71})$$

A small positive numerical floor is used only to avoid logarithms of nonpositive roundoff values inside the blend window. The resulting dimensionless reflectivity is exactly Parratt below the selected window, exactly the scaled structure branch above the window, and smooth through the crossover.

The subscript stitch labels the replacement axial strength after this crossover. The dimensionless blend is converted to the axial structural strength before detector weighting. In the per- dq_z convention of the optical calculation,

$$S_{q,\text{stitch}}(Q_{\text{loc}}) = \frac{Q_{\text{loc}}^2 S_{q,00}(0)}{A_{\text{opt}}} R_{\text{blend}}(Q_{\text{loc}}/Q_c). \quad (\text{S72})$$

Above the crossover this recovers the finite-stack strength evaluated at L_{film} . The stitch supplies the replacement axial structural strength and enters the detector weighting once. Equation (S22) supplies the conversion to the per- dL density used in Sec. S4.

For the joint comparison, the low- Q_{loc} interval and the 003 neighborhood share one normalization. The measured detector profile, the Parratt branch, the finite-stack branch, and the combined curve must share one declared background treatment and one intensity scale. The 003 peak serves as the higher- Q_{loc} structural reference. Its sensitivity to wavelength spread can be examined with Eqs. (S27) and (S28). The model controls described above retain the radial extension for a monochromatic source. The contribution of wavelength broadening to the measured feature remains to be established. Figure S6 reserves the physical comparison on that common scale.

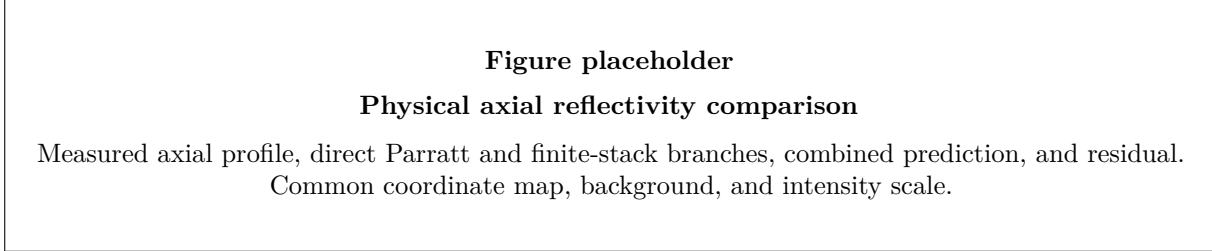


Figure S6. Placeholder for the physical axial comparison. The low-angle interval and diffraction region are to be calculated from one declared structural and optical state with uncertainty analysis.

S6 Additional ordered-film comparisons

The main text displays the nominal 5° Bi_2Se_3 and Bi_2Te_3 conditions. This section reserves the direct measured/calculated comparisons needed to assess the ordered-film model. Images at 5° , 10° , and 15° entered the staged refinement, so comparisons at those conditions assess agreement with the fitted observations.

For each material, the comparison record identifies the exposure, structural input, source distribution, mosaic state, optical assumptions, background, intensity scale, and finite integration regions. Measured and calculated profiles use the same coordinate map and normalization from Sec. S4. A residual is defined as measured minus calculated intensity. Error bars describe the propagated measurement uncertainty and are distinct from numerical convergence errors and parameter uncertainty.

Both detector sides of each retained off-axis family are kept identifiable. Off-axis profiles use L_{proj} , while an axial profile may retain detector 2θ when that is the measured coordinate. A complete comparison preserves negative background-subtracted measurements and records any excluded interval. Figures S7 and S8 identify the corresponding comparisons.

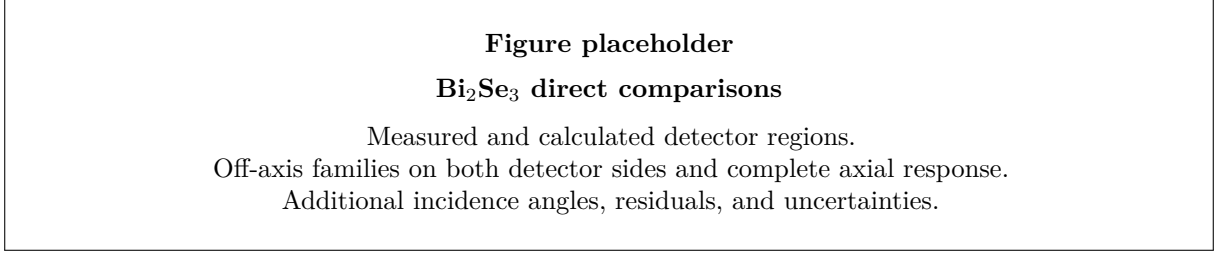


Figure S7. Placeholder for Bi₂Se₃ direct measured/calculated comparisons at the declared incident conditions. Each profile must identify its source state, integration region, normalization, uncertainty, and residual.

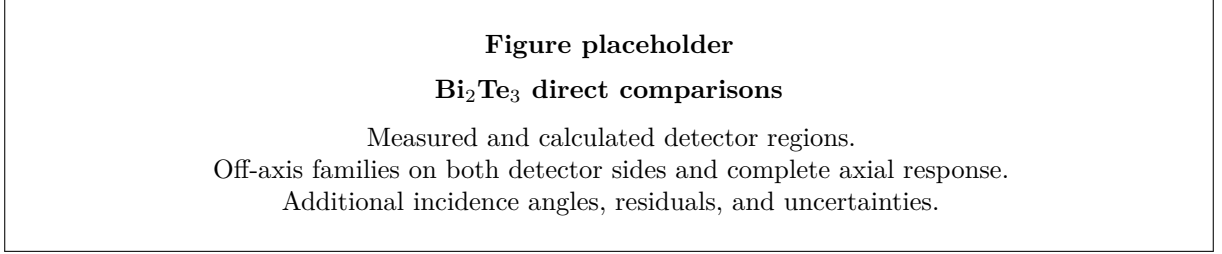


Figure S8. Placeholder for Bi₂Te₃ direct measured/calculated comparisons at the declared incident conditions. Each profile must identify its source state, integration region, normalization, uncertainty, and residual.

S7 PbI₂ stacking-disorder model

This section gives the PbI₂-specific state construction and the finite-stack intensity used in the main text. The general strategy—propagating layer-pair probabilities with a transition matrix—follows the established theory of one-dimensionally disordered crystals.[5–8] The particular six states, allowed interface events, and fault templates below are the model assumptions introduced for the present PbI₂ analysis.

S7.1 Trilayer amplitudes and six-state basis

As in the main text, $\rho \in \{0, 1, 2\}$ labels the basal registry and $\sigma \in \{+, -\}$ labels trilayer orientation. Here σ is a discrete state label; the mosaic width σ in Sec. S3 is a separate, continuous parameter. The basal registries are

$$0 = (0, 0), \quad 1 = \left(\frac{1}{3}, \frac{2}{3}\right), \quad 2 = \left(\frac{2}{3}, \frac{1}{3}\right). \quad (\text{S73})$$

Taking the Pb plane as the local origin and writing the Pb–I vertical offset as $\eta = \Delta z_{\text{I–Pb}}/c$, the two orientation-dependent trilayer amplitudes are

$$F_+(h, k, L) = f_{\text{Pb}} + f_{\text{I}} \exp\left\{2\pi i \left[\frac{2h+k}{3} + L\eta\right]\right\} + f_{\text{I}} \exp\left\{2\pi i \left[\frac{h+2k}{3} - L\eta\right]\right\}, \quad (\text{S74})$$

$$F_-(h, k, L) = f_{\text{Pb}} + f_{\text{I}} \exp\left\{2\pi i \left[\frac{2h+k}{3} - L\eta\right]\right\} + f_{\text{I}} \exp\left\{2\pi i \left[\frac{h+2k}{3} + L\eta\right]\right\}. \quad (\text{S75})$$

Atomic form factors retain their dependence on $Q = |\mathbf{G}_{hk}(L)|$ and source wavelength λ_s . These arguments are suppressed in the matrix expressions below. The two amplitudes satisfy $F_-(h, k, L) = F_+(h, k, -L)$.

For the forward registry step $\mathbf{s} = (1/3, 2/3)$, define

$$\omega_{hk} = \exp[2\pi i(hs_x + ks_y)] = \exp\left[\frac{2\pi i}{3}(h + 2k)\right]. \quad (\text{S76})$$

The ordered state basis and its scattering-amplitude vector are

$$\mathcal{S} = [0F_+, 1F_+, 2F_+, 0F_-, 1F_-, 2F_-], \quad \mathbf{g} = \begin{pmatrix} F_+ & \omega_{hk}F_+ & \omega_{hk}^2F_+ & F_- & \omega_{hk}F_- & \omega_{hk}^2F_- \end{pmatrix}^\top. \quad (\text{S77})$$

Because $\omega_{hk}^3 = 1$, registry 2 is also one backward step from registry 0.

S7.2 Direction-resolved interface law

The five mutually exclusive interface events are

$$\boldsymbol{\theta} = (a, b_+, b_-, d_+, d_-), \quad a + b_+ + b_- + d_+ + d_- = 1. \quad (\text{S78})$$

The entries of $\boldsymbol{\theta}$ are event probabilities. Here a is the probability of retaining both registry and orientation; the basal lattice constant denoted by a in Sec. S2 is a different quantity. The a event preserves registry and orientation. The b_+ and b_- events shift the registry by +1 and -1 while preserving orientation. The d_+ and d_- events combine an orientation change with the two handed shift rules. Every allowed orientation flip includes a registry shift. For a d_+ event, an $F_+ \rightarrow F_-$ step advances the registry and the corresponding $F_- \rightarrow F_+$ step returns it. The directions are interchanged for d_- . This reversal is what makes a pure d_+ or d_- process a deterministic two-layer 4H repeat.

Let

$$S_+ = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad S_- = S_+^{-1} = S_+^\top. \quad (\text{S79})$$

The matrices S_+ and S_- shift the registry forward and backward, respectively. Let I_m denote the m -dimensional identity matrix. With rows denoting the current state and columns the next state, define

$$A = aI_3 + b_+S_+ + b_-S_-, \quad (\text{S80})$$

$$B = d_+S_+ + d_-S_-, \quad (\text{S81})$$

$$C = d_+S_- + d_-S_+. \quad (\text{S82})$$

Here A is the orientation-preserving block, and B and C are the two orientation-changing blocks. The full six-state transition matrix is

$$T_6 = \begin{pmatrix} A & B \\ C & A \end{pmatrix}. \quad (\text{S83})$$

For example, the row for the state $0F_+$ is $(a, b_+, b_-, 0, d_+, d_-)$. Every row of T_6 sums to one.

The deterministic parent-event vectors, in the order (a, b_+, b_-, d_+, d_-) , are

$$\mathbf{e}_{2H} = (1, 0, 0, 0, 0), \quad \mathbf{e}_{4H} = (0, 0, 0, 1, 0), \quad \mathbf{e}_{6H} = (0, 1, 0, 0, 0). \quad (\text{S84})$$

The opposite-handed 4H and 6H parents are obtained by selecting d_- and b_- instead.

S7.3 Exact rod-dependent reduction

The three registry-shift matrices are diagonal in the discrete Fourier basis. In the sector selected by the scattering phase ω_{hk} , the 6×6 transition reduces exactly to

$$M(\omega_{hk}) = \begin{pmatrix} a + b_+\omega_{hk} + b_-\omega_{hk}^{-1} & d_+\omega_{hk} + d_-\omega_{hk}^{-1} \\ d_+\omega_{hk}^{-1} + d_-\omega_{hk} & a + b_+\omega_{hk} + b_-\omega_{hk}^{-1} \end{pmatrix}. \quad (\text{S85})$$

All registry states are retained. Equation (S85) is one Fourier block of Eq. (S83). Setting the registry phase to unity gives the orientation-population matrix

$$P = M(1) = \begin{pmatrix} a + b_+ + b_- & d_+ + d_- \\ d_+ + d_- & a + b_+ + b_- \end{pmatrix}. \quad (\text{S86})$$

The matrix $M(\omega_{hk})$ carries complex scattering phases, while P is a row-stochastic probability matrix.

S7.4 Finite-stack intensity

Let d be the separation between neighboring trilayer centers and $\xi = \exp(iq_z d)$ the corresponding vertical phase, with $q_z = 2\pi L/c$. Define

$$\mathbf{f} = \begin{pmatrix} F_+ \\ F_- \end{pmatrix}, \quad D_{F^*} = \text{diag}(F_+^*, F_-^*). \quad (\text{S87})$$

If π_0^{ori} is the initial orientation row vector, then the orientation population at trilayer n is

$$\pi_n^{\text{ori}} = \pi_0^{\text{ori}} P^n. \quad (\text{S88})$$

The vector $\mathbf{f}$ collects the two orientation amplitudes, and D_{F^*} is their diagonal complex-conjugate matrix. The superscript *ori* labels orientation probabilities; each row vector π_n^{ori} sums to one. For integer layer separation $1 \leq \Delta < N$, the phase-weighted correlation between trilayers n and $n + \Delta$ is

$$K_{n,\Delta} = \pi_n^{\text{ori}} D_{F^*} M(\omega_{hk})^\Delta \mathbf{f}. \quad (\text{S89})$$

For one fixed parent-rich population $j \in \{2H, 4H, 6H\}$, the main-text rod intensity uses the per-trilayer convention $S_{hk}^{(j)} = \langle I_N^{(j)} \rangle / N$, where $I_N^{(j)}$ is the intensity of the complete N -trilayer stack before the disorder average. Population labels on $F_\pm$, the matrices, and $K_{n,\Delta}$ are suppressed in this single-population derivation. For a finite stack of N trilayers,

$$S_{hk}^{(j)}(L, \lambda_s) = \frac{1}{N} \sum_{n=0}^{N-1} \pi_n^{\text{ori}} \begin{pmatrix} |F_+|^2 \\ |F_-|^2 \end{pmatrix} + \frac{2}{N} \text{Re} \left[\sum_{\Delta=1}^{N-1} \xi^\Delta \sum_{n=0}^{N-1-\Delta} K_{n,\Delta} \right]. \quad (\text{S90})$$

The connection to the main-text pair-correlation notation is

$$C_{hk}^{(j)}(\Delta; L, \lambda_s) = \frac{1}{N - \Delta} \sum_{n=0}^{N-1-\Delta} K_{n,\Delta}, \quad 1 \leq \Delta < N. \quad (\text{S91})$$

The first term in Eq. (S90) is $S_{\text{self},hk}^{(j)}(L, \lambda_s)$. Since $\xi^\Delta = e^{iq_z d \Delta}$, grouping the pair sum with this definition gives the main-text finite-stack expression directly.

The first term contains the N self contributions and the second contains all distinct trilayer pairs. The calculation evaluates these finite sums directly, retaining end effects and avoiding singular closed forms when an eigenvalue approaches unity. Equivalent matrix-power and recursive formulations are standard for planar-fault diffraction.[9]

S7.5 Polytype-rich fault templates and population mixture

For each parent-rich population,

$$\boldsymbol{\theta}_j(\epsilon_j) = (1 - \epsilon_j)\mathbf{e}_j + \epsilon_j\mathbf{q}_j, \quad j \in \{2H, 4H, 6H\}. \quad (\text{S92})$$

Here $\mathbf{e}_j$ is the deterministic parent-event vector, $\mathbf{q}_j$ is the probability vector for a fault event, and $0 \leq \epsilon_j \leq 1$ is its probability. The equal-alternative templates used in the present implementation are

$$\mathbf{q}_{2H} = \left(0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right), \quad (\text{S93})$$

$$\mathbf{q}_{4H} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}\right), \quad (\text{S94})$$

$$\mathbf{q}_{6H} = \left(\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right). \quad (\text{S95})$$

Consequently, ϵ_j is exactly the probability that an interface in population j follows one of the four alternative events. Its numerical value is conditional on the chosen $\mathbf{q}_j$ and is interpreted together with that template.

The parent-rich populations are mutually incoherent, with nonnegative weights w_j that sum to one. Their structural intensities combine using the main-text notation:

$$S_{hk}(L, \lambda_s) = \sum_{j \in \{2H, 4H, 6H\}} w_j S_{hk}^{(j)}(L, \lambda_s), \quad w_{2H} + w_{4H} + w_{6H} = 1. \quad (\text{S96})$$

The detector/profile prediction, background, scale and optional finite-profile kernel are specified separately in Sec. S8.2. The core parameterization has three disorder magnitudes and two independent population weights. The stack length N , scale, and any fixed or refined finite-profile and instrumental broadening terms are separate. When three independent disorder magnitudes remain unconstrained, the constrained starting model is $\epsilon_{2H} = \epsilon_{4H} = \epsilon_{6H}$.

S7.6 Deterministic and phase-insensitive checks

Table S2 gives the zero-disorder cycles.

Table S2. Deterministic PbI₂ parent cycles from the initial state $0F_+$. Each row specifies the unit-probability interface event, its registry/orientation cycle, and the corresponding reduced transfer matrix. The phase ω_{hk} is defined in Eq. (S76); I_2 is the two-dimensional identity matrix.

Parent	Nonzero event	State cycle from $0F_+$	Reduced matrix
2H	$a = 1$	$0F_+ \rightarrow 0F_+$	I_2
4H	$d_+ = 1$	$0F_+ \rightarrow 1F_- \rightarrow 0F_+$	$\begin{pmatrix} 0 & \omega_{hk} \\ \omega_{hk}^{-1} & 0 \end{pmatrix}$
6H	$b_+ = 1$	$0F_+ \rightarrow 1F_+ \rightarrow 2F_+ \rightarrow 0F_+$	$\omega_{hk} I_2$

For 4H, $M(\omega_{hk})^2 = I_2$. For 6H, $M(\omega_{hk})^3 = I_2$ because $\omega_{hk}^3 = 1$. The $d_- = 1$ and $b_- = 1$ limits generate the opposite-handed cycles.

For $h = k = 0$, $\omega_{hk} = 1$ and $F_+ = F_- = F$. Because $P^\Delta \mathbf{1} = \mathbf{1}$, where $\mathbf{1} = (1, 1)^\top$, Eq. (S89) becomes $K_{n,\Delta} = |F|^2$ for every allowed transition law. Equation (S90) then reduces to the normalized Laue factor

$$S_{00}^{(j)}(L, \lambda_s) = \frac{|F|^2}{N} \left| \sum_{n=0}^{N-1} \xi^n \right|^2 = \frac{|F|^2}{N} \frac{\sin^2(N\vartheta/2)}{\sin^2(\vartheta/2)}, \quad \vartheta = q_z d, \quad \xi = e^{i\vartheta}. \quad (\text{S97})$$

Thus the strong $r = 0$ oscillations are finite-thickness fringes. More generally, $h + 2k = 3m$, with integer m , gives $\omega_{hk} = 1$ and $F_+ = F_-$, so the same independence from the transition law holds for every such rod. This includes the $r = 0$ and $r = 3$ families. Their intensities are unchanged by the included registry and orientation changes at fixed stack length and trilayer structure. Figure S9 reserves the direct numerical checks of these limits. Figure S10 separates the effects of faults within a stack from changes in population weights.

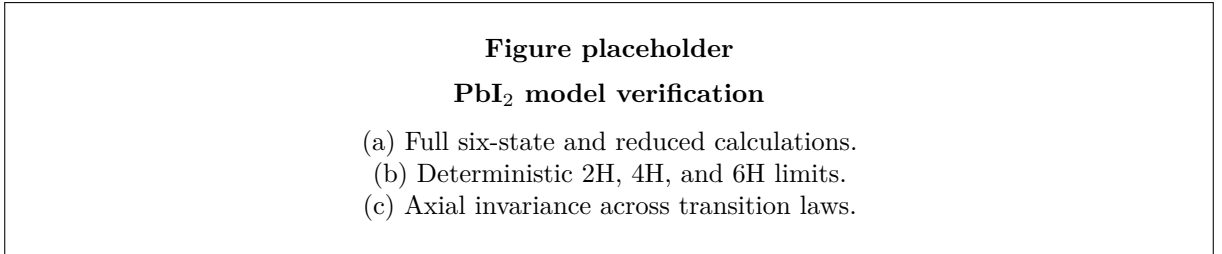


Figure S9. Placeholder for verification of the PbI₂ model. Direct arrays will compare the full and reduced expressions, independent deterministic-parent sums, and the phase-insensitive axial limit.

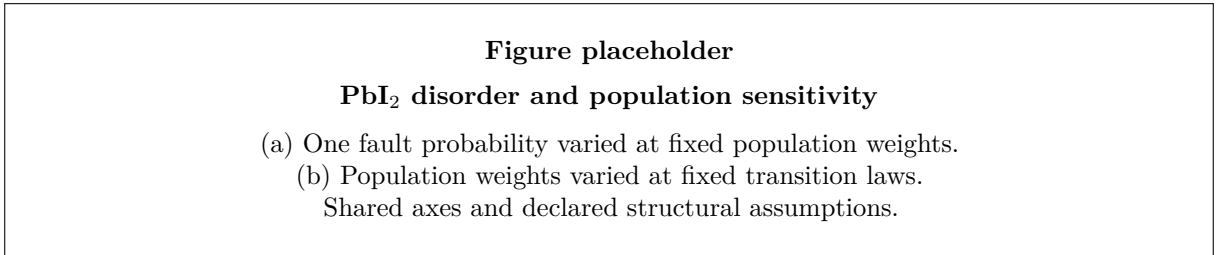


Figure S10. Placeholder for direct PbI₂ sensitivity calculations. Separate controlled panels distinguish within-stack fault probabilities from incoherent population weights.

S8 Calibration, refinement, and uncertainty

The main-text workflow separates parameter groups because several effects can change the same peak position, width, or intensity. This section defines the stage-specific observables and the propagation of uncertainty through the shared detector integration.

S8.1 Parameter groups and calibration

1. **Beam characterization (direct beam).** Direct-beam spot positions and shapes at several detector distances constrain the incident-beam widths (w_x, w_z) and Gaussian divergences ($\Delta\theta_x, \Delta\theta_z$) that set the instrument-limited broadening.
2. **Detector geometry calibration (3D powder standard).** Dark-subtracted Debye–Scherrer rings from a polycrystalline hBN calibrant constrain detector distance, detector tilts (β_D, γ_D), in-plane detector rotation χ_D , and beam center (x_0, y_0).
3. **Sample alignment and goniometer misalignment.** Indexed Bragg-peak centers constrain sample alignment with the detector parameters fixed. The positional objective minimizes detector-space residuals between measured and calculated peak positions spanning both specular and off-specular regions. Here θ_i is grazing incidence, χ is the in-plane sample rotation, and z_B and z_S are the beam and sample height coordinates in the laboratory frame. The adjustable quantities are $(\theta_i, \chi, z_B, z_S)$ and the goniometer-axis misalignment rotations $(\alpha_{\text{gon}}, \psi)$. Here α_{gon} denotes the alignment angle labeled α in the main-text workflow table; the subscript distinguishes it from the crystallite tilt α in Sec. S3. Finite film dimensions (W, H) enter only when finite support is selected, while thickness t enters the absorption model. The retained ordered-film geometry uses the unbounded in-plane support described in Sec. S1.
4. **Mosaicity/profile refinement.** With geometry fixed, selected local detector regions or local $I(\phi_f)$ profiles after center alignment constrain mosaicity. The shared profile model is a wrapped two-component angular distribution with a Gaussian-like core and a Lorentzian-like broad component. The two components have independent widths. Off-specular arcs mainly inform the narrow core, while off-condition axial intensity is sensitive to the broad component together with bandwidth, divergence, alignment, and background. The parameterization yields component widths and a broad-component weight, but attribution and necessity require the controlled no-tail and re-optimized Gaussian-only comparisons at fixed remaining parameters and after refitting, respectively.
5. **Ordered-structure refinement.** With geometry and mosaicity fixed, the ordered intensity objective compares the model with background-subtracted integrated counts in selected detector-space Bragg regions of interest (ROIs) from multiple dark-subtracted images. The corresponding prediction integrates the calculated detector density over those same ROIs. The objective permits one nuisance scale per image and physically constrained ordered-structure parameters.
6. **Disorder refinement.** After the ordered baseline is established, stacking-disorder parameters enter through fixed-in-plane-momentum rod profiles reported versus the film-normal momentum. The measured profile is formed from caked signal and normalization fields by summing signal and normalization over the selected rod mask before division. Geometry,

detector calibration, beam terms, mosaicity, lattice constants, and the ordered baseline remain fixed. Only stacking-disorder parameters and declared scale/background terms are refined.

All detector images are dark-subtracted before fitting. Any additional flat-field, polarization, or detector-efficiency correction enters the common intensity convention for the compared images. The calculated detector density already includes the pixel solid-angle factor. Applying that factor again would duplicate the correction. Stage-specific uncertainty could be estimated by repeated constrained refits, resampling calibrant points for detector geometry, indexed peak lists for sample alignment, local reflection regions for mosaicity, accepted Bragg regions for ordered intensity, and selected rod bins for disorder. The spread of the accepted solutions would then quantify uncertainty within each stage and contribute to uncertainty in subsequent stages.

The hBN calibration requires a reproducible record of the selected rings, masks, fitted geometry, positional residuals, and uncertainty. Likewise, a final parameter table must distinguish fixed structural inputs from refined values and link each value to the calculation state used for the comparisons in Sec. S6.

S8.2 Ordered-intensity and disorder objectives

The detector-space forward model predicts a continuous detector-coordinate density and its finite-region integrals, while the inverse problems use stage-specific observables. In the ordered-structure stage, the measured quantity for ROI i in exposure e is the background-subtracted detector count

$$y_{ei} = \sum_{(c_D, r_D) \in \Omega_{ei}} [I_e^{\text{meas}}(c_D, r_D) - b_{ei}(c_D, r_D)],$$

Here I_e^{meas} is the native pixel count in exposure e , Ω_{ei} is the selected detector-space Bragg ROI and b_{ei} is its local background count model. The sum runs over pixel centers within that ROI. The corresponding prediction is the detector-coordinate density integrated over the same region,

$$\mu_{ei} = \int_{\Omega_{ei}} I_{\text{det},e}^{\text{calc}}(c_D, r_D) dc_D dr_D,$$

or the equivalent deterministic sum of native pixel masses. Geometry, source terms, and mosaicity are fixed before this stage. One nuisance scale per image accounts for exposure and flux differences. This defines a multi-image ordered-intensity objective restricted to the selected Bragg ROIs.

The PbI₂ disorder refinement is restricted to the final-stage selected profiles. The upstream beam, detector, sample alignment, mosaic/profile parameters, lattice constants, and ordered PbI₂ baseline are fixed before this stage starts. The measured selected-rod profile is the signal-to-normalization ratio in Eq. (S48). Its reciprocal-space mask and detector-side selection remain fixed during this stage.

The parameter vector Θ_{dis} contains the released fault probabilities and independent population weights, together with N if stack length is refined. The model profile is evaluated along the same selected branch. A scalar convolution approximates the detector response when its variation along that branch can be represented by a fixed kernel. In the common intrinsic- L coordinate used for that approximation,

$$I_{\nu}^{\text{calc}}(\Theta_{\text{dis}}) = B_{\text{bg}}(x_{\nu}) + a_{\text{sc}} \left[K_{\text{prof}} * \sum_{j \in \{2H, 4H, 6H\}} w_j S_{hk}^{(j)}(L, \lambda_s; \epsilon_j, N) \right]_{\nu}. \quad (\text{S98})$$

Here ν labels a retained profile bin, x_ν is its displayed coordinate, B_{bg} is the low-order background, a_{sc} is a common intensity scale, and K_{prof} is the fixed finite-profile kernel. The binary $*$ denotes convolution in intrinsic L . The bracket denotes evaluation and integration over the same selected bin as the measurement. The source wavelength and other upstream inputs are fixed in this compact single-state expression; a source ensemble is accumulated with the weights w_s before the same profile extraction. The structural functions $S_{hk}^{(j)}$ are those of Sec. S7. Kernel widths are fixed by the upstream resolution and mosaic state or declared before the fit; peak areas remain governed by the structural model.

The scale and background arguments are suppressed in I_ν^{calc} ; they remain explicit in the objective. The disorder objective uses only the retained profile-bin set $\mathcal{V}$:

$$\Phi_{\text{dis}}(\Theta_{\text{dis}}, a_{\text{sc}}, B_{\text{bg}}) = \sum_{\nu \in \mathcal{V}} \mathcal{L} \left(\frac{I_\nu^{\text{ROI}} - I_\nu^{\text{calc}}(\Theta_{\text{dis}})}{\sigma_{I,\nu}} \right). \quad (\text{S99})$$

Here I_ν^{ROI} is the measured ratio defined in Eq. (S48), $\sigma_{I,\nu}^2 = \text{Var}(I_\nu^{\text{ROI}})$ is its intensity variance, and $\mathcal{L}$ is the declared loss function. For standardized residual v , the choice $\mathcal{L}(v) = v^2$ gives the usual pointwise least-squares objective; a robust loss defines a different objective. Correlated profile bins require the covariance treatment below. The fixed detector geometry and ordered model determine the known Bragg locations. Fitting independent peak amplitudes before the stacking model would absorb the diffuse between-peak intensity into preprocessing.

S8.3 Coordinate and intensity uncertainty

The transformed-coordinate uncertainty has two parts: uncertainty in the coordinate label and uncertainty in the rebinned intensity. The coordinate-label uncertainty follows from the finite detector-pixel footprint and from uncertainty in the detector calibration. For square detector pixels with pitch p_{pix} and reference detector distance D , the leading flat-detector terms are

$$\sigma_{2\theta, \text{pix}}(2\theta) = \frac{p_{\text{pix}} \cos^2(2\theta)}{\sqrt{12} D}, \quad \sigma_{\phi_f, \text{pix}}(2\theta) = \frac{p_{\text{pix}}}{\sqrt{12} D \tan(2\theta)}. \quad (\text{S100})$$

The radial term is finite and largest near the direct beam. The azimuthal term scales as $1/\tan(2\theta)$, so azimuth is poorly conditioned near the direct-beam direction, where 2θ approaches zero.

For the $D = 75$ mm and $p_{\text{pix}} = 300 \mu\text{m}$ geometry used in the uncertainty calculation, the maximum radial coordinate uncertainty is approximately 0.066° . The azimuthal coordinate uncertainty falls below 1° above $2\theta \approx 3.8^\circ$, below 0.5° above $2\theta \approx 7.5^\circ$, and below 0.25° above $2\theta \approx 14.9^\circ$. More generally, the threshold for a target azimuthal width w_{ϕ_f} is

$$2\theta_{\text{crit}} = \arctan \left(\frac{p_{\text{pix}}}{\sqrt{12} D w_{\phi_f}} \right), \quad (\text{S101})$$

with w_{ϕ_f} expressed in radians.

Additional calibration uncertainty can be propagated with the usual Jacobian form. If the uncertain geometry parameters are collected in $\mathbf{g}_{\text{cal}}$ with covariance matrix Σ_{cal} , the row Jacobians are $J_{2\theta} = \partial(2\theta)/\partial \mathbf{g}_{\text{cal}}$ and $J_{\phi_f} = \partial \phi_f / \partial \mathbf{g}_{\text{cal}}$. The propagated variances are

$$\sigma_{2\theta, \text{cal}}^2 = J_{2\theta} \Sigma_{\text{cal}} J_{2\theta}^T, \quad \sigma_{\phi_f, \text{cal}}^2 = J_{\phi_f} \Sigma_{\text{cal}} J_{\phi_f}^T. \quad (\text{S102})$$

If the displayed bin-center label is also treated as uncertain, the bin widths $\Delta(2\theta)$ and $\Delta\phi_f$ contribute

$$\sigma_{2\theta,\text{bin}}^2 = \frac{[\Delta(2\theta)]^2}{12}, \quad \sigma_{\phi_f,\text{bin}}^2 = \frac{(\Delta\phi_f)^2}{12}. \quad (\text{S103})$$

These coordinate terms describe the uncertainty of the transformed axis value. They are separate from intensity uncertainty.

The intensity uncertainty is inherited from the detector through the same overlap weights used to construct the caked image and the projected profile. If v_p is the detector-pixel variance and the detector-pixel variances are independent, then

$$\text{Var}(I_\nu^{\text{ROI}}) = \frac{\sum_p a_{\nu p}^2 v_p}{\left(\sum_p a_{\nu p} n_p\right)^2}. \quad (\text{S104})$$

If overlapping interpolation weights are used, neighboring profile points can be statistically correlated. For profile bins ν and ν' , the corresponding covariance is

$$\text{Cov}(I_\nu^{\text{ROI}}, I_{\nu'}^{\text{ROI}}) = \frac{\sum_p a_{\nu p} a_{\nu' p} v_p}{\left(\sum_p a_{\nu p} n_p\right) \left(\sum_p a_{\nu' p} n_p\right)}. \quad (\text{S105})$$

Thus the same weights that define the projected profile also propagate the detector-pixel variance. This expression assumes fixed normalization and masks and independent input-pixel variances. Correlated background or calibration errors require additional covariance terms. Off-diagonal covariance terms describe correlations between neighboring projected points.

The displayed coordinate for a projected point can be represented by the same finite-band weighting. For example,

$$\bar{L}_{\text{proj},\nu} = \frac{\sum_b \mu_{\nu b} N_b L_{\text{proj},b}}{\sum_b \mu_{\nu b} N_b}, \quad \sigma_{L_{\text{proj},\nu}}^2 = \frac{\sum_b \mu_{\nu b} N_b (L_{\text{proj},b} - \bar{L}_{\text{proj},\nu})^2}{\sum_b \mu_{\nu b} N_b}, \quad (\text{S106})$$

Here $\bar{L}_{\text{proj},\nu}$ is the weighted coordinate mean, $\sigma_{L_{\text{proj},\nu}}^2$ is its finite-band variance, and $L_{\text{proj},b}$ is the coordinate at caked bin b . These terms precede the addition of calibration, incidence-angle, wavelength, divergence, and mosaic contributions. This width describes the finite-bin projection. Counting-statistical intensity uncertainty is propagated separately through the detector-pixel variances.

These projection uncertainties are coupled to model-parameter correlations. Peak positions are sensitive to detector distance, beam center, detector tilt, sample height, and incidence angle. Profile widths are sensitive to both beam divergence and the Gaussian mosaic core. Off-condition profiles are sensitive to the broad mosaic component together with bandwidth, divergence, alignment, and background. Near-critical intensity additionally depends on the optical response and is excluded from evidence for the broad component. Relative integrated intensities are sensitive to scale, footprint, absorption, thickness, and structure-factor parameters. The staged workflow separates these effects by fixing or constraining the parameters tied to peak position before fitting profile widths, and by fixing the geometry and mosaic state before interpreting ordered intensities.

These expressions give four practical consequences. First, the 2θ - ϕ_f maps and Q_R -selected Q_z profiles are controlled projections that inherit the detector image's finite resolution and statistical information. Second, transformed-coordinate uncertainty is concentrated at low angles

near the direct beam. Third, intensity variance is propagated through the same overlap weights used for caking and profile extraction. Fourth, the strongest remaining systematic coupling occurs near low L , where direct-beam background, bandwidth, divergence, incidence angle, and optical response overlap. That near-critical region is excluded from evidence for the broad mosaic component.

S8.4 Numerical evaluation and convergence

The physical sequence is source sampling, sample acceptance, internal propagation, orientation-weighted Ewald selection, exit propagation, detector mapping, and finite-region integration. The forward roots in Sec. S4.1 and inverse detector evaluation describe the same compatible scattering states. Numerical source sampling integrates the incident-beam distribution, while the separate mosaic law specifies crystallite orientations.

Source-state count and detector integration order control different approximations. A convergence comparison holds the physical parameters, masks, background and count scale fixed while changing one numerical setting at a time. Agreement is assessed for the actual detector regions and profiles used in the manuscript, including narrow peaks and low-angle regions. Establishing convergence requires quantitative agreement under these changes in numerical settings. Figure S11 reserves the corresponding numerical comparison.

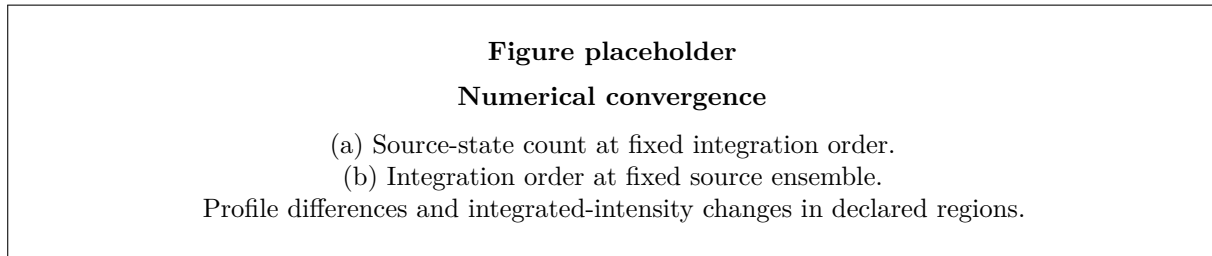


Figure S11. Placeholder for numerical convergence of the detector observables. The intended comparison reports source sampling and finite-area quadrature separately, with fixed physical inputs and explicit tolerances.

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